

Regular SQ

Seq	Op	ST Addr	ST Data	Note
1	ST	A	xxx	older ST
2	ST	B	xxx	no-alias
4	spm_cp	A	a (spm addr)	bridge entry
...	...	...	...	...

Regular LQ

Seq	Op	LD Addr	Note
7	LD	A	younger LD
8	LD	C	no-alias
...	...	...	...

SPM SQ

Seq	Op	ST Addr	ST Data	Note
4	spm_cp	a	xxx	install pending
6	spm_st	b	xxx	no-alias
...	...	...	...	...

SPM LQ

Seq	Op	LD Addr	Note
3	spm_ld	a	older load
5	spm_ld	a	younger load
...	...	...	...

→ forwarding

--→ no forwarding, no squash

(a) spm_cp on the regular side

RAW hazard: hold/squash on conflict

(b) spm_cp on the SPM side

Regular SQ

Seq	Op	ST Addr	ST Data	Note
1	ST	A	xxx	older ST
2	ST	B	xxx	no-alias
4	spm_wb	A	a (spm addr)	bridge entry
...	...	...	...	...

Regular LQ

Seq	Op	LD Addr	Note
5	LD	A	younger LD
9	LD	C	no-alias
...	...	...	...

SPM SQ

Seq	Op	ST Addr	ST Data	Note
3	spm_st	a	xxx	older ST
7	spm_st	a	xxx	younger ST
...	...	...	...	...

SPM LQ

Seq	Op	LD Addr	Note
4	spm_wb	a	wb pending
8	spm_ld	d	no-alias
...	...	...	...

RAW: wait until coherent install (no forwarding)

(c) spm_wb on the regular side

WAR hazard: stall on conflict

(d) spm_wb on the SPM side